Learning Dynamics for High-Speed Tractor-Trailer Control

Aaron Chen*[1], David Shen*[1], Taekyung Kim*[2], Kaleb Ben Naveed[2], Dimitra Panagou[2,3]

***Abstract*— Model predictive control (MPC) is widely used for autonomous vehicles, but model mismatch can degrade its performance. We study learning-based MPC for tractor-trailer systems under uncertain vehicle and environmental dynamics. Tractor-trailer systems are especially unstable during reverse motion, and their dynamics depend on difficult-to-measure factors, including road friction and suspension behavior. We present a learned dynamics model for tractor-trailer control and introduce a reduced-order dynamics model with Fiala tire forces as an analytic baseline. We compare our fully learned dynamics model within a model predictive path integral (MPPI) controller against an analytic prior and several baseline controllers. In the main closed-loop tracking experiments across four simulated environments, learned MPPI achieves mean speeds closer to the target magnitudes and lower root-mean-square (RMS) hitch angles than analytic-model MPPI at every tested setpoint, with higher completion rates in seven of eight reverse configurations. [Project Page] [Code]**

I. INTRODUCTION

Autonomous vehicle control has advanced [1], [2], but reliable control under uncertain dynamics remains challenging [3]. Model predictive control (MPC) uses a dynamics model to optimize a finite-horizon control sequence subject to input and state constraints. The controller relies on these predictions to compare the costs of candidate trajectories and to anticipate violations of the constraints. Errors in the predicted motion can therefore affect which control inputs are selected and degrade closed-loop performance [3], [4].

These challenges are particularly pronounced in tractor-trailer systems, whose coupled dynamics are nonlinear [5]. With an actuated tractor and a passive trailer, the vehicle combination is underactuated and particularly unstable during reverse motion [6]. Tractor-trailer dynamics are also sensitive to variables that are difficult to measure accurately, including load transfer, road friction, road slope, and suspension dynamics. Reverse tracking therefore requires the controller to maintain progress along the track while regulating the relative motion of the tractor and trailer.

Recent approaches to handling dynamical and environmental uncertainty have focused on integrating learned dynamics into MPC controllers, reducing the reliance on a fixed analytic dynamics prior [3], [7]–[10].

A related approach to handling uncertainty is policy adaptation. Rapid Motor Adaptation (RMA) conditions a base policy on a latent representation of the operating conditions [11]. The policy is trained in simulation with privileged environmental information. At deployment, an adaptation module infers the latent representation from recent state-action history, while the base policy weights remain fixed. This approach has also been extended to bipedal robots [12]. Safe Deep Policy Adaptation combines an adaptive reinforcement learning policy with learned dynamics and a control barrier function (CBF) filter [13]. Our approach uses recent history to condition a dynamics predictor, which MPPI uses to evaluate candidate control sequences.

Learned dynamics can replace an analytic predictor [9], [14] or provide a residual correction to an analytic prior [15]. Gaussian processes (GPs) provide predictive uncertainty estimates and are useful in low-data regimes [10]. Neural dynamics models offer an alternative for learning from large offline datasets [3], [7], [16]. Conditioning predictions on recent state-action histories can provide context about latent operating conditions [9], [17].

Recent tractor-trailer methods address forward and reverse parking using barrier-rate-guided MPPI [18], navigation using learned geometric distance estimates within MPPI [19], and safety-critical trajectory planning for general N-trailer systems [20]. High-speed studies have used analytic models for lateral stabilization and forward trajectory tracking, including active trailer steering [21] and tube MPC for robustness to model uncertainty and disturbances [22]. Learned vehicle dynamics have also been used to improve robustness to state-estimation error and environmental variation, including in towing tasks [17]. We focus on explicitly predicting and regulating tractor-trailer articulation during high-speed reverse tracking under uncertain dynamics.

In this work, we apply learned dynamics to high-speed control of tractor-trailer systems under both system and environmental uncertainty. Our main contributions are:

- We derive a reduced-order tractor-trailer dynamics model with Fiala tire forces, providing an analytic control baseline.
- We integrate a compact history-conditioned dynamics model with MPPI for high-speed forward and reverse tractor-trailer control under uncertain vehicle and environmental dynamics.
- We compare five-second trajectory predictions from the learned model and analytic model with and without privileged friction information.
- We evaluate closed-loop tracking across four simulated environments and compare selected predictors with different history lengths on a separate Snow + Slope segment.

*These authors contributed equally to this work.

[1]Department of Electrical Engineering and Computer Science, [2]Department of Robotics, [3]Department of Aerospace Engineering, University of Michigan, Ann Arbor, MI, 48109, USA {aach, dyshen, taekyung, kbnaveed, dpanagou}@umich.edu

II. PRELIMINARIES

A. Notation

The sets of natural and real numbers are denoted by $\mathbb{N}$ and $\mathbb{R}$, respectively. We denote the normal distribution with mean μ and covariance Σ by $\mathcal{N}(\mu, \Sigma)$. For a sequence $\{\boldsymbol{a}_n\}$, we write $\boldsymbol{a}_{p:q} = (\boldsymbol{a}_p, \ldots, \boldsymbol{a}_q)$, with both endpoints included. We use t for the current controller time and τ for time within the prediction horizon.

B. System Dynamics

Consider a discrete-time nonlinear system

$$\boldsymbol{x}_{t+1} = f(\boldsymbol{x}_t, \boldsymbol{u}_t) \tag{1}$$

where $\boldsymbol{x}_t \in \mathcal{X} \subseteq \mathbb{R}^n$ is the state at time $t \in \mathbb{N}$, $\boldsymbol{u}_t \in \mathcal{U} \subseteq \mathbb{R}^m$ is the control input at time t, and $f : \mathcal{X} \times \mathcal{U} \to \mathcal{X}$ is the discrete-time dynamics. Only part of the system state is directly observed, so we partition $\boldsymbol{x}_t = (\boldsymbol{s}_t, \boldsymbol{z}_t)$, where $\boldsymbol{s}_t \in \mathcal{S} \subseteq \mathbb{R}^s$ is observed and $\boldsymbol{z}_t \in \mathcal{Z} \subseteq \mathbb{R}^{n-s}$ is not directly observed.

C. Model Predictive Path Integral Control

Model predictive path integral (MPPI) control is a sampling-based method for controlling nonlinear systems [7], [23]. At control step t, let $\hat{\boldsymbol{u}}_{t:t+T-1}$ be the nominal control sequence. MPPI samples K control sequences by drawing $\boldsymbol{v}_\tau^{(k)} \sim \mathcal{N}(\hat{\boldsymbol{u}}_\tau, \Sigma)$ independently for $\tau = t, \ldots, t+T-1$ and $k = 1, \ldots, K$. Each sequence is simulated under an internal dynamics function $\tilde{f} : \mathcal{S} \times \mathcal{U} \to \mathcal{S}$. For the learned predictor, $\tilde{f}$ additionally depends on the state-action history described in Section III-B; this dependence is suppressed here for readability. For each rollout, the predicted state is initialized at the current controller time t, with $\tilde{\boldsymbol{s}}_t^{(k)} = \boldsymbol{s}_t$. Denoting the predicted state-action pair for the k-th rollout at time τ as $(\tilde{\boldsymbol{s}}_\tau^{(k)}, \boldsymbol{v}_\tau^{(k)})$, the k-th rollout's state prediction for time $\tau + 1$ is calculated as

$$\tilde{\boldsymbol{s}}_{\tau+1}^{(k)} = \tilde{f}(\tilde{\boldsymbol{s}}_\tau^{(k)}, \boldsymbol{v}_\tau^{(k)}), \quad \tau = t, \ldots, t+T-1. \tag{2}$$

The rollout cost is

$$S^{(k)} = \sum_{\tau=t}^{t+T-1} \left[C(\tilde{\boldsymbol{s}}_\tau^{(k)}, \boldsymbol{v}_\tau^{(k)}) + \gamma\, \hat{\boldsymbol{u}}_\tau^\top \Sigma^{-1} \left(\boldsymbol{v}_\tau^{(k)} - \hat{\boldsymbol{u}}_\tau \right) \right], \tag{3}$$

where the second term is the importance-sampling correction, scaled by the tunable parameter γ. The normalized sampling weights are

$$\eta = \sum_{k=1}^{K} \exp\left(-\frac{1}{\lambda} \left(S^{(k)} - \min_{1 \le j \le K} S^{(j)} \right) \right) \tag{4}$$

$$w^{(k)} = \frac{1}{\eta} \exp\left(-\frac{1}{\lambda} \left(S^{(k)} - \min_{1 \le j \le K} S^{(j)} \right) \right) \tag{5}$$

where $\lambda > 0$ is the sampling temperature. The control sequence is updated as $\hat{\boldsymbol{u}}_\tau \leftarrow \sum_{k=1}^{K} w^{(k)} \boldsymbol{v}_\tau^{(k)}$. Its first input is applied to the plant, and the remaining sequence is shifted to initialize the next control cycle.

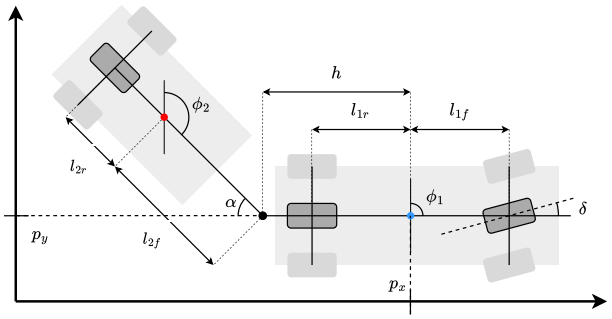


Fig. 1: Illustration of our tractor-trailer model. The tractor is approximated by a bicycle model and the trailer by a single axle connected to the tractor at a common hitch point.

D. Problem Statement

Given an offline dataset of state-action trajectories, we seek to learn an observed-state dynamics model that takes in a fixed-length state-action history for high-speed forward and reverse tractor-trailer control. We evaluate whether using this model within MPPI improves prediction and closed-loop performance under unobserved vehicle and environmental dynamics.

III. METHODOLOGY

A. Tractor-trailer Dynamics

We derive a planar tractor-trailer dynamics model with a configuration similar to that of [24]. The model jointly describes longitudinal, lateral, and yaw dynamics and uses Fiala tire forces [25]. It serves as our analytic baseline.

We approximate the tractor by a bicycle model and the trailer by a single axle, as shown in Fig. 1. Subscripts 1 and 2 denote the tractor and trailer, respectively. For body i, ϕ_i is its yaw angle, m_i its mass, and I_{zi} its yaw moment of inertia about its center of mass. The distances l_{1f} and l_{1r} run from the tractor's center of mass to its front and rear axles; l_{2f} and l_{2r} run from the trailer's center of mass to the hitch and its modeled axle. The distance from the tractor's center of mass to the hitch is h. We define the hitch (articulation) angle as $\alpha := \phi_1 - \phi_2$, and δ denotes the front-wheel steering angle.

The tractor's center-of-mass position is $\boldsymbol{p}_1 := (p_{1x}, p_{1y})$, and v_{1x}, v_{1y} are its longitudinal and lateral body-frame velocities. We define $\boldsymbol{f}_{1f} := (f_{1xf}, f_{1yf})$ as the tractor's front tire forces in the front-wheel frame, $\boldsymbol{f}_{1r} := (f_{1xr}, f_{1yr})$ as the tractor's rear tire forces in the body frame, and $\boldsymbol{f}_{2r} := (f_{2xr}, f_{2yr})$ as the trailer's tire forces in the trailer frame.

Let $c := \cos\alpha$, $s := \sin\alpha$, $\bar{m} := m_1 + m_2$, $\rho := m_2 l_{2f}$, and let R_θ denote rotation by θ, so that R_α maps tractor-frame components to trailer-frame components. We write $\boldsymbol{v}_i := (v_{ix}, v_{iy})$ for body i's body-frame velocity and $\boldsymbol{\nu} := (v_{1x}, v_{1y}, \dot{\phi}_1, \dot{\phi}_2)$.

Let $\boldsymbol{f}_h$ be the force the trailer exerts on the tractor at the hitch, expressed in the tractor frame. The Newton–Euler

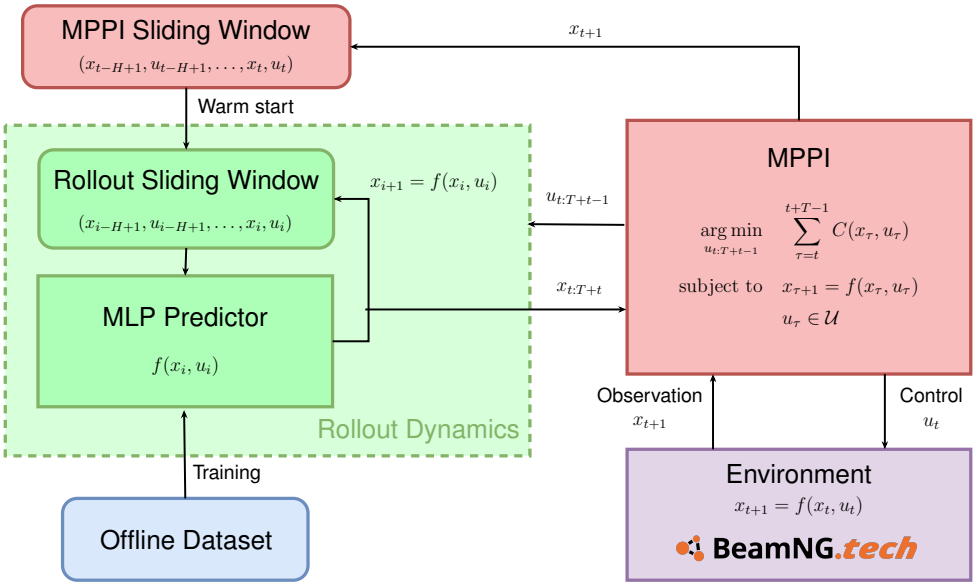


Fig. 2: Overview of our controller architecture. MPPI uses a multilayer perceptron (MLP) trained offline to predict dynamic-state and actuator rates from a sliding window of state-action pairs. These predictions are integrated recursively to generate trajectory rollouts.

equations in each body frame are

$$\begin{aligned} m_1\left(\dot{\boldsymbol{v}}_1 + \dot{\phi}_1 J \boldsymbol{v}_1\right) &= \boldsymbol{f}_{1r} + R_\delta \boldsymbol{f}_{1f} + \boldsymbol{f}_h \\ I_{z1}\ddot{\phi}_1 &= -l_{1r} f_{1yr} + l_{1f}(R_\delta \boldsymbol{f}_{1f})_y - h f_{hy} \quad (6) \\ m_2\left(\dot{\boldsymbol{v}}_2 + \dot{\phi}_2 J \boldsymbol{v}_2\right) &= \boldsymbol{f}_{2r} - R_\alpha \boldsymbol{f}_h \\ I_{z2}\ddot{\phi}_2 &= -l_{2r} f_{2yr} - l_{2f}(R_\alpha \boldsymbol{f}_h)_y\,, \quad (7) \end{aligned}$$

with $J := \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$. The two bodies share a common hitch point, so the trailer velocity is determined by the tractor state and $\dot{\phi}_2$,

$$\boldsymbol{v}_2 = R_\alpha \begin{bmatrix} v_{1x} \\ v_{1y} - h\dot{\phi}_1 \end{bmatrix} + \begin{bmatrix} 0 \\ -l_{2f}\dot{\phi}_2 \end{bmatrix}. \quad (8)$$

Differentiating (8) yields

$$\dot{\boldsymbol{v}}_2 = R_\alpha \begin{bmatrix} \dot{v}_{1x} \\ \dot{v}_{1y} - h\ddot{\phi}_1 \end{bmatrix} + \dot{\alpha}\, J R_\alpha \begin{bmatrix} v_{1x} \\ v_{1y} - h\dot{\phi}_1 \end{bmatrix} + \begin{bmatrix} 0 \\ -l_{2f}\ddot{\phi}_2 \end{bmatrix}. \quad (9)$$

Equations (6) and (7) cannot be solved sequentially: $\dot{\boldsymbol{v}}_1$ depends on $\boldsymbol{f}_h$, which by (7) depends on $\dot{\boldsymbol{v}}_2$, which by (9) depends on $\dot{\boldsymbol{v}}_1$.

Substituting (8) and (9) into (6)–(7), eliminating $\boldsymbol{f}_h$, and collecting the four unknown accelerations yields

$$M(\alpha)\,\dot{\boldsymbol{\nu}} = \boldsymbol{b}_{\text{tire}} + \boldsymbol{b}_{\text{cor}}, \quad (10)$$

where

$$M(\alpha) = \begin{bmatrix} \bar{m} & 0 & 0 & -\rho s \\ 0 & \bar{m} & -m_2 h & -\rho c \\ 0 & -m_2 h & I_{z1} + m_2 h^2 & \rho h c \\ -\rho s & -\rho c & \rho h c & I_{z2} + m_2 l_{2f}^2 \end{bmatrix}. \quad (11)$$

Writing $\boldsymbol{f}_1 := \boldsymbol{f}_{1r} + R_\delta \boldsymbol{f}_{1f}$ for the total tractor tire force, $\hat{\boldsymbol{f}}_2 := R_\alpha^\top \boldsymbol{f}_{2r}$ and $\tilde{\boldsymbol{v}}_2 := R_\alpha^\top \boldsymbol{v}_2$ for the trailer force and velocity expressed in the tractor frame, and $g := \rho\dot{\alpha}\dot{\phi}_2$, we have

$$\boldsymbol{b}_{\text{tire}} = \begin{bmatrix} f_{1x} + \hat{f}_{2x} \\ f_{1y} + \hat{f}_{2y} \\ -l_{1r} f_{1yr} + l_{1f}(R_\delta \boldsymbol{f}_{1f})_y - h\hat{f}_{2y} \\ -(l_{2f} + l_{2r}) f_{2yr} \end{bmatrix}, \quad (12)$$

$$\boldsymbol{b}_{\text{cor}} = \begin{bmatrix} \dot{\phi}_1(m_1 v_{1y} + m_2 \tilde{v}_{2y}) + gc \\ -\dot{\phi}_1(m_1 v_{1x} + m_2 \tilde{v}_{2x}) - gs \\ m_2 h \dot{\phi}_1 \tilde{v}_{2x} + hgs \\ \rho v_{2x} \dot{\phi}_1 \end{bmatrix}. \quad (13)$$

Lateral tire forces are modeled using the Fiala brush model,

$$f_y = \begin{cases} -C_c \tan\beta + \dfrac{C_c^2 |\tan\beta| \tan\beta}{3\mu f_z} - \dfrac{C_c^3 \tan^3\beta}{27\mu^2 f_z^2}, & |\beta| < \beta_{\text{sl}}, \\ -\mu f_z \operatorname{sgn}\beta, & \text{otherwise}, \end{cases} \quad (14)$$

where C_c is the cornering stiffness, f_z is the normal load, and μ is the tire–road friction coefficient. The saturation slip angle is $\beta_{\text{sl}} = \arctan(3\mu f_z / C_c)$. We define the wheel slip angle as $\beta = \operatorname{atan2}(v_\perp, \max(|v_\parallel|, \epsilon))$, where $v_\parallel$ and $v_\perp$ are wheel-frame longitudinal and lateral velocities and $\epsilon > 0$ regularizes the low-speed denominator. This convention makes the lateral force oppose lateral slip.

The accelerations follow by solving (10): $\dot{\boldsymbol{\nu}} = M(\alpha)^{-1}(\boldsymbol{b}_{\text{tire}} + \boldsymbol{b}_{\text{cor}})$.

B. Controller Architecture

1) Dataset Collection: For the standard learned model, we collect an offline dataset of simulated trajectories using analytic-model MPPI. Trajectories span different friction levels and target velocities, with noise injected into the MPPI control inputs. To increase coverage of tire-force responses, we add forward-driving trajectories with sinusoidal weaving. We further augment the dataset with trajectories collected by

an intermediate learned controller near the vehicle handling limits.

2) Model Training: Our standard model is a multilayer perceptron (MLP) that predicts observed-state dynamics from $H = 4$ state-action pairs; Section IV-C compares other history lengths. For brevity, we write $v_x = v_{1x}$ and $v_y = v_{1y}$. The model takes input

$$(\sin\alpha, \cos\alpha, v_x, v_y, \dot{\phi}_1, \dot{\phi}_2, \hat{u}_\delta, \hat{u}_a, u_\delta, u_a)_{t-H+1:t}$$

and predicts the finite-difference rates

$$(\dot{v}_x, \dot{v}_y, \ddot{\phi}_1, \ddot{\phi}_2, \dot{\hat{u}}_\delta, \dot{\hat{u}}_a)_t.$$

Here $u_\delta, u_a \in [-1, 1]$ are normalized steering and longitudinal actuation commands, and $\hat{u}_\delta, \hat{u}_a$ are the corresponding observed actuator states. Positive u_a commands throttle and negative u_a commands braking; $\hat{u}_a$ is realized throttle minus brake, rather than physical acceleration. The normalized steering command u_δ is distinct from the front-wheel angle δ used in the analytic model. Targets are differences between consecutive observed velocity, yaw-rate, and actuator states divided by Δt. Including both commanded and observed actuator states gives the predictor information about how the vehicle has responded to recent control inputs. Velocity, yaw-rate, and actuator states are updated using forward Euler, while position and orientation are propagated kinematically. Thus, "fully learned" refers to the predicted dynamic and actuator channels. The MLP has four hidden layers of 128 neurons each. Inputs and targets are standardized using training-split means and standard deviations.

3) Controller Overview: As depicted in Fig. 2, each MPPI rollout starts from the measured state-action history and propagates the learned model recursively. Its history window is updated with predicted states and sampled controls, so each candidate control sequence generates its own predicted history. The costs of the K rollouts determine the control sequence update, and the first command is applied to the plant. The resulting observation and the applied command then update the measured history for the next control cycle. In the main tracking experiments, we initialize this history with H steps of analytic-model MPPI.

4) Cost Function: The cost promotes target-velocity tracking and penalizes track-boundary and hitch-angle violations. The violation penalty c_{viol} is

$$\begin{aligned} c_{\text{viol}} &= c_{\text{track}} + c_{\text{jackknife}} \\ c_{\text{track}} &= \max\left(0, |e_{\text{lat}}| - 0.45W_{\text{track}} + 0.1\right) \\ c_{\text{jackknife}} &= \max(0, |\alpha| - \alpha_{\max}) \end{aligned} \tag{15}$$

where e_{lat} is the signed deviation of the tractor's center of mass from the centerline of the track, $\alpha_{\max}$ is the maximum hitch angle, and W_{track} is the width of the track. When $|\alpha| > \alpha_{\max}$, we consider the tractor-trailer to have jackknifed. Additionally, we define

$$c_{\text{vel}} = |v^\star - v_{\text{track}}| \tag{16}$$

$$c_{\text{center}} = e_{\text{lat}}^2 \tag{17}$$

$$c_{\text{hitch}} = \alpha^2 \tag{18}$$

where $v^\star$ is the signed target velocity and v_{track} is the signed velocity along the track centerline. Combining (15)–(18), the stage cost at rollout offset $j = \tau - t \in \{0, \ldots, T-1\}$ is

$$C = 0.99^j \cdot 10^7 c_{\text{viol}} + k_v c_{\text{vel}} + k_c c_{\text{center}} + k_a c_{\text{hitch}}, \tag{19}$$

where k_v, k_c, and k_a are penalty weights.

C. Simulation Environments

We evaluate the controllers in BeamNG.tech [26], whose internal plant dynamics are unavailable to the controllers. The control inputs are the normalized commands (u_δ, u_a), and the observations are $(p_{1x}, p_{1y}, \phi_1, \phi_2, v_{1x}, v_{1y}, \dot{\phi}_1, \dot{\phi}_2, \hat{u}_\delta, \hat{u}_a)$. The tractor-trailer follows a predefined racetrack at high target speeds in forward and reverse motion, providing a stress test of dynamics prediction. The track follows the general layout of the Circuit de Barcelona-Catalunya. Unobserved plant states and parameters $\boldsymbol{z}_t$ include vehicle pitch and roll, suspension states, local road friction and roughness, and powertrain behavior.

D. Controllers

We evaluate the following controllers:

i) *Controller 1*: An MPPI controller using the learned dynamics model described in Section III-B.
ii) *Controller 2*: An MPPI controller with the analytic prior described in Section III-A.
iii) *Controller 3*: A hybrid controller that uses a linear quadratic regulator (LQR) [27] for steering and proportional–integral–derivative (PID) control [28] for longitudinal actuation. The LQR controller operates on a linearization of the analytic dynamics model at the $\alpha = 0$ operating point and assumes $v_y = 0$ and constant v_x.
iv) *Controller 4*: A gradient-based MPC controller that uses IPOPT [29] with direct shooting under the analytic dynamics model. The cost is a smoothed version of (19).
v) *Controller 5*: A gradient-based model predictive contouring control (MPCC) controller [30] using our analytic dynamics model.

IV. RESULTS

We address three research questions:

1. **Q1**: Does learned MPPI improve forward and reverse tracking relative to analytic-model MPPI across friction and terrain variations?
2. **Q2**: How do learned and analytic dynamics compare in multi-step prediction, and does privileged friction information close the gap?
3. **Q3**: How does closed-loop performance compare among selected learned predictors with history lengths $H \in \{1, 4, 8\}$?

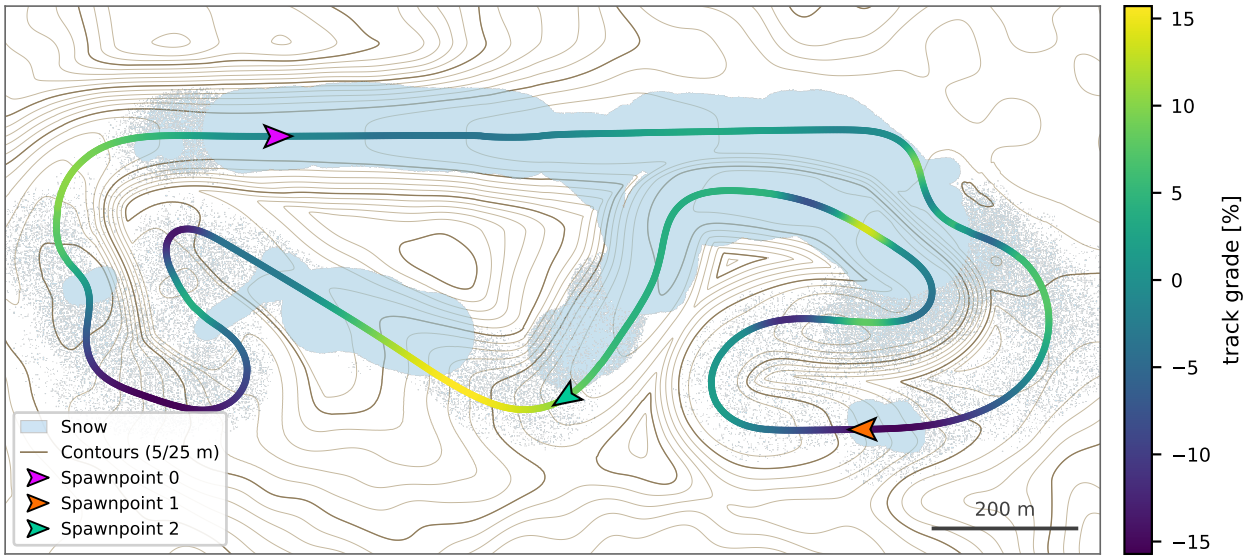


Fig. 3: Top-down view of the Snow + Slope test environment, based on the Circuit de Barcelona-Catalunya layout. Snow patches and terrain changes are applied to the base asphalt track. The snow pattern matches the Snow environment, and the terrain profile matches the Slope environment.

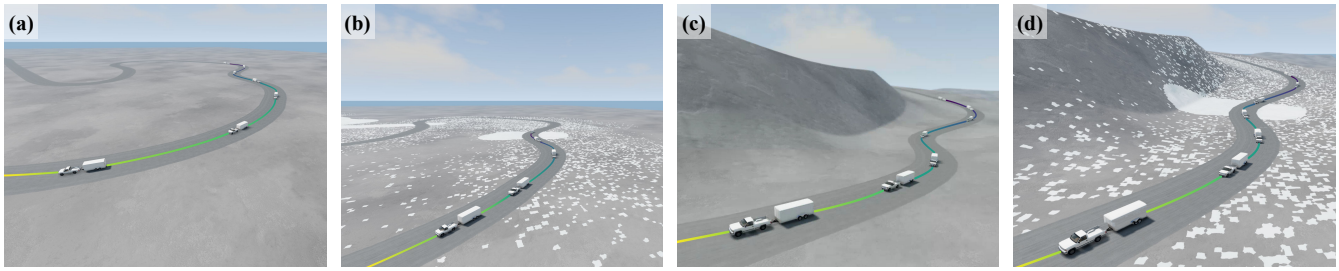


Fig. 4: Representative reverse-driving trajectories of Controller 1 in (a) Asphalt, (b) Snow, (c) Slope, and (d) Snow + Slope. White patches indicate snow-covered regions with lower friction; the colored curve shows the trajectory over time.

A. Closed-Loop Tracking

We evaluate forward and reverse tracking in four environments. Figure 3 shows the combined snow and terrain modifications; Fig. 4 shows representative reverse-driving trajectories in each environment.

1. *Asphalt*: A flat track with uniformly high-friction asphalt.
2. *Snow*: A flat asphalt track with irregularly distributed snow patches that vary the local friction.
3. *Slope*: A uniformly high-friction track with elevation changes and banking.
4. *Snow + Slope*: The same asphalt-snow pattern as in the Snow scenario, with the same terrain pattern as the Slope scenario.

Table I gives the MPPI settings for these main tracking experiments. Table II summarizes performance in the four scenarios.

1) Asphalt Scenario: Controller 1 is the only controller to complete all nine 500-step runs at each of the three setpoints, $v^\star \in \{-30, -50, 80\}\,\mathrm{km/h}$. It achieves the lowest RMS hitch angle at both reverse setpoints. In forward motion, Controller 3 achieves a lower RMS hitch angle but completes fewer runs. Controller 1 also achieves mean speeds closer to the target magnitudes than Controller 2 at every setpoint.

TABLE I: MPPI settings for the main tracking experiments. Sampling covariance is expressed in normalized command coordinates (u_δ, u_a).

Parameter	Forward	Reverse
k_v	1×10^2	5×10^1
k_c	1×10^0	5×10^1
k_a	7×10^2	1×10^2
λ (temperature)	150	100
diag(Σ)	$(7 \times 10^{-2}, 0.2)$	$(2 \times 10^{-2}, 0.2)$
K (samples)	500	500
T (horizon steps)	80	65
γ	0.0	0.0
Δt (s)	0.05	0.05

At $-50\,\mathrm{km/h}$, for example, Controller 1 achieves $\bar{v} = 47.2\,\mathrm{km/h}$, $\bar{\alpha}_{\mathrm{rms}} = 1.75°$, and nine completions, compared with $22.7\,\mathrm{km/h}$, $6.70°$, and six completions for Controller 2.

2) Snow Scenario: Controller 1 completes all nine Snow runs at each setpoint, with mean speeds closer to the target magnitudes than Controller 2. It has no jackknife or stall failures despite its higher reverse speeds; Controller 2 stalls in four of nine runs at each reverse setpoint. Controller 3 has higher RMS hitch angles in reverse but a slightly lower forward value ($2.36°$ vs. $2.72°$). Controller 4 has higher RMS hitch angles and fewer completed steps than Controller 1 at every setpoint. Controller 5 has the fewest completed steps at each setpoint and the fewest completions overall.

TABLE II: Closed-loop tracking across four environments, with nine runs per configuration. Controllers are defined in Section III-D. Bold hitch-angle values are the lowest for each velocity setpoint and environment.

Controller	$v^\star$	Asphalt				Snow				Slope				Snow + Slope			
	(km/h)	$\bar{v}$ (km/h)	$\bar{\alpha}_{\mathrm{rms}}$ (deg)	$\bar{N}$ (steps)	C/J/S/O/F (out of 9)	$\bar{v}$ (km/h)	$\bar{\alpha}_{\mathrm{rms}}$ (deg)	$\bar{N}$ (steps)	C/J/S/O/F (out of 9)	$\bar{v}$ (km/h)	$\bar{\alpha}_{\mathrm{rms}}$ (deg)	$\bar{N}$ (steps)	C/J/S/O/F (out of 9)	$\bar{v}$ (km/h)	$\bar{\alpha}_{\mathrm{rms}}$ (deg)	$\bar{N}$ (steps)	C/J/S/O/F (out of 9)
1: Learned MPPI [7]	-30	29.1	**1.14**	**500**	9/0/0/0/0	29.1	**1.18**	**500**	9/0/0/0/0	25.4	**1.03**	455	8/0/1/0/0	20.4	**10.82**	386	6/0/3/0/0
	-50	47.2	**1.75**	**500**	9/0/0/0/0	46.0	**1.82**	**500**	9/0/0/0/0	42.1	**3.06**	498	8/1/0/0/0	30.4	**15.41**	413	4/0/5/0/0
	80	75.4	2.57	**500**	9/0/0/0/0	72.6	2.72	**500**	9/0/0/0/0	70.7	2.67	**500**	9/0/0/0/0	69.1	2.83	**500**	9/0/0/0/0
2: Analytic MPPI [7]	-30	19.7	5.25	392	5/0/4/0/0	19.6	7.82	403	5/0/4/0/0	22.3	2.86	498	8/0/1/0/0	16.7	11.47	365	4/0/5/0/0
	-50	22.7	6.70	448	6/0/3/0/0	22.8	8.49	417	5/0/4/0/0	22.5	6.48	457	6/0/3/0/0	18.1	15.86	321	2/1/6/0/0
	80	71.1	3.00	**500**	9/0/0/0/0	68.7	3.26	**500**	9/0/0/0/0	67.1	3.19	**500**	9/0/0/0/0	64.8	3.66	**500**	9/0/0/0/0
3: LQR–PID [27], [28]	-30	30.8	4.80	495	8/0/0/1/0	30.6	5.36	439	6/0/0/3/0	28.5	10.02	446	6/0/0/3/0	24.0	16.56	377	3/6/0/0/0
	-50	45.1	14.41	452	6/1/0/2/0	48.8	8.97	407	6/3/0/0/0	46.9	10.58	396	6/3/0/0/0	37.1	16.91	352	3/3/0/3/0
	80	74.8	**2.45**	436	6/0/0/3/0	72.4	**2.36**	442	6/0/0/3/0	71.7	**2.52**	**500**	9/0/0/0/0	70.7	**2.53**	**500**	9/0/0/0/0
4: Shooting MPC [29]	-30	28.4	3.11	**500**	9/0/0/0/0	23.5	20.22	373	5/1/3/0/0	19.9	16.32	478	7/2/0/0/0	16.7	20.79	338	4/0/5/0/0
	-50	32.2	54.93	107	0/9/0/0/0	31.9	39.96	244	0/6/0/3/0	26.3	28.09	293	3/3/3/0/0	23.5	30.59	292	3/3/3/0/0
	80	71.3	12.28	471	5/1/0/3/0	69.1	9.16	468	6/3/0/0/0	69.5	8.13	490	6/0/0/3/0	66.7	11.80	475	6/0/0/3/0
5: MPCC [30]	-30	22.0	26.57	301	0/3/2/4/0	21.9	39.16	261	0/3/3/3/0	18.1	30.46	374	1/0/7/0/1	14.1	31.79	275	0/1/6/2/0
	-50	39.3	14.74	261	0/3/1/4/1	30.2	24.95	203	0/2/3/4/0	37.3	17.98	352	1/3/1/3/1	25.5	25.63	202	0/1/3/5/0
	80	68.0	6.06	400	4/0/0/5/0	66.1	4.82	407	4/0/0/4/1	64.5	6.73	379	3/0/0/5/1	64.4	5.74	426	4/0/0/5/0

Each configuration comprises three seeds at each of three locations ($n = 9$). Our notation is as follows: $v^\star$ is the signed velocity setpoint; negative values denote reverse motion. $\bar{v}$ and $\bar{\alpha}_{\mathrm{rms}}$ are the arithmetic means of the per-run mean speed magnitudes and RMS hitch angles, respectively, computed after the first 100 control steps. $\bar{N}$ is the mean number of completed control steps out of 500. All means include unsuccessful runs and weight each run equally. Bold $\bar{N}$ entries indicate that all nine runs completed 500 steps. C/J/S/O/F gives the numbers of completed, jackknifed, stalled, off-track, and solver-failed runs, respectively. Completion denotes reaching the 500-step (25-second) test horizon. A stall is defined by $|v_x| < 0.5\,\mathrm{m/s}$ with no track progress for 2 s.

3) Slope Scenario: In the Slope environment, Controller 1 completes eight of nine runs at each reverse setpoint, compared with all nine on Asphalt and Snow. The two reverse terminations consist of one stall at $-30\,\mathrm{km/h}$ and one jackknife at $-50\,\mathrm{km/h}$. It still achieves mean speeds closer to the target magnitudes and lower RMS hitch angles than Controller 2 at every setpoint. It completes at least as many reverse runs and has lower reverse RMS hitch angles than every baseline. In forward motion, Controller 3 matches its completion rate with a slightly higher mean speed and lower RMS hitch angle.

4) Snow + Slope Scenario: Controller 1 has the most reverse completions and the lowest reverse RMS hitch angles on Snow + Slope, although stalls limit its completion rate. This combined environment also has the largest reduction in learned MPPI's reverse completions: 10 of 18 runs, compared with 18 of 18 on Snow and 16 of 18 on Slope. It is the only controller with no jackknife failures across the three setpoints. Controller 3 achieves a higher mean speed and slightly lower RMS hitch angle in forward driving, but completes fewer reverse runs, with more jackknife or off-track failures.

5) Summary Across Environments: Across the eight reverse configurations in Table II, Controller 1 completes 62 of 72 runs, compared with 41 of 72 for Controller 2. The corresponding stall counts are nine and 30, while each controller has one jackknife failure. Controller 1 also achieves higher mean speeds and lower RMS hitch angles in every reverse configuration. Both MPPI variants complete all 36 forward runs. LQR–PID has lower forward RMS hitch angles than learned MPPI in all four environments, with fewer completions on Asphalt and Snow. These results answer **Q1**: the completion advantage is concentrated in reverse motion, with 21 fewer stalls and the same number of jackknife failures.

The learned predictor receives no terrain-map input, unlike the terrain-aware model in [16], and slope and banking are absent from its training data. It transfers to the tested terrain profiles, but combined snow and terrain variation still causes stalls: Controller 1 completes six of nine reverse runs at $-30\,\mathrm{km/h}$ and four of nine at $-50\,\mathrm{km/h}$.

B. Dynamics Prediction Error

We compare the multi-step dynamics prediction accuracy of the following models:

1. The fixed-friction analytic prior, with $\mu = 1$.
2. The friction-aware analytic prior, given the measured tractor and trailer contact-friction sequence.
3. The fully learned model with $H = 4$.

We select one five-second segment per direction and environment from BeamNG.tech trajectories collected with learned MPPI. All predictors start from the same measured state and replay the same 100 recorded commands at $\Delta t = 0.05\,\mathrm{s}$ without state resets. The learned model uses four measured state-action pairs for initialization. Replaying a fixed command sequence allows us to compare the models' accumulated prediction errors without differences in the control actions chosen during the rollout. The friction-aware prior receives the recorded contact-friction sequence, rather than a friction map queried along its predicted path. Figure 5 shows the Snow + Slope rollouts, absolute wrapped hitch-angle error, and absolute longitudinal velocity error. Table III reports position and wrapped hitch-angle RMSE over the 100 predicted states, excluding the common initial state.

During reverse driving, the learned model achieves the lowest position RMSE in all four environments and the lowest hitch-angle RMSE in three of four. On Asphalt, the fixed-friction prior has lower hitch-angle RMSE (96.5° versus 101.1°). Figure 5(e) shows slower initial growth of the

TABLE III: Position (e_p) and wrapped hitch-angle (e_α) RMSE for one five-second segment per direction and environment (meters and degrees, respectively). Bold values indicate the lowest RMSE for each environment, direction, and metric.

		Prior		Friction-aware prior		Learned	
Direction	Environment	e_p	e_α	e_p	e_α	e_p	e_α
Reverse	Asphalt	5.8	**96.5**	7.4	113.0	**4.2**	101.1
	Snow	7.0	130.1	4.9	107.9	**4.0**	**88.2**
	Slope	7.7	132.6	8.2	134.2	**2.2**	**78.7**
	Snow + Slope	8.1	131.5	7.3	111.6	**2.1**	**83.4**
Forward	Asphalt	7.3	93.4	6.6	95.3	**1.4**	**2.1**
	Snow	**7.3**	82.3	11.9	83.8	10.8	**16.0**
	Slope	3.5	80.4	**2.7**	79.2	5.7	**1.1**
	Snow + Slope	8.4	85.7	14.7	84.0	**7.5**	**8.7**

learned model's hitch-angle error, although its error becomes large later in the five-second horizon.

During forward driving, the learned model achieves the lowest hitch-angle RMSE in all four environments. Position RMSE is mixed: the learned model performs best on Asphalt and Snow + Slope, while the fixed-friction prior performs best on Snow and the friction-aware prior performs best on Slope. Across the eight selected segments, the friction-aware prior has lower position RMSE than the fixed-friction prior in four cases and lower hitch-angle RMSE in four cases. Providing the recorded friction sequence therefore has a mixed effect on the analytic predictions and does not consistently close the gap to the learned model (**Q2**). These comparisons do not establish how much of the remaining error comes from unmodeled vehicle behavior and how much comes from other environmental effects. The five-second diagnostic horizon also exceeds the main MPPI horizons of $4.0\,\mathrm{s}$ forward and $3.25\,\mathrm{s}$ reverse; the large late reverse errors show the limits of open-loop prediction despite the closed-loop tracking results above. Because hitch error is wrapped, a late decrease after a large trailer rotation need not indicate recovery.

C. History-Length Comparison

We compare selected predictors with $H \in \{1, 4, 8\}$ on a separate Snow + Slope segment, using three seeds at one start per direction (Table IV). The $H = 1$ model uses fixed-friction training; $H = 4$ and $H = 8$ use mixed-friction data. All runs start from rest with eight shared physical warm-up steps; each network retains its last H measured pairs.

This comparison uses $K = 1500$, $T = 80/75$ for forward/reverse driving, $\lambda = 100$, $\mathrm{diag}(\Sigma) = (0.01, 0.2)$, $\gamma = 0$, and $\Delta t = 0.05\,\mathrm{s}$. The weights (k_v, k_c, k_a) are $(20, 50, 100)$ forward and $(50, 50, 200)$ reverse. All candidates use these numerical MPPI settings. Both the starting segment and tuning differ from the main experiments, so completion rates should not be compared directly with Table II.

The $H = 1$ model completes no runs in either direction; all three reverse runs end in jackknifing within the first 100 control steps. Both $H = 4$ and $H = 8$ complete all reverse runs, with similar RMS hitch angles of $2.84°$ and $2.87°$. The $H = 4$ mean speed is also closer to the target magnitude than that of $H = 8$.

TABLE IV: Closed-loop history comparison on Snow + Slope.

Predictor	$v^\star$ (km/h)	$\bar{v}$ (km/h)	$\bar{\alpha}_{\mathrm{rms}}$ (deg)	$\bar{N}$ (steps)	C/J/S/O/F (out of 3)
$H = 1$	-30	15.8	29.29	98	0/3/0/0/0
	80	48.0	22.27	278	0/2/0/1/0
$H = 4$	-30	30.0	**2.84**	**500**	3/0/0/0/0
	80	51.6	29.29	345	0/3/0/0/0
$H = 8$	-30	30.9	2.87	**500**	3/0/0/0/0
	80	64.0	**6.07**	**500**	3/0/0/0/0
Analytic	-30	18.9	13.78	393	2/0/1/0/0
	80	53.3	20.00	356	0/2/0/1/0

Each model has three seeds at one start location per direction. $\bar{v}$ and $\bar{\alpha}_{\mathrm{rms}}$ are equal-run means after the first 100 steps (the whole run if it ends within 100 steps); $\bar{N}$ is the mean duration out of 500 steps (25 s). C/J/S/O/F counts completed/jackknifed/stalled/off-track/solver-failed runs. Bold indicates minimum RMS hitch angle per direction, or all runs reaching 500 steps.

Their forward outcomes differ: $H = 4$ jackknifes in all three runs, whereas $H = 8$ completes all three and has the lowest forward RMS hitch angle. Its mean speed of $64.0\,\mathrm{km/h}$ remains below the $80\,\mathrm{km/h}$ target. Forward hitch RMS does not decrease monotonically with H: the $H = 4$ value exceeds that of $H = 1$. The analytic reference completes two reverse runs and no forward runs. Although its forward mean speed and RMS hitch angle are better than those of $H = 4$, neither completes a forward run. These results answer **Q3** for the selected predictors; differences in their training data prevent attributing the outcomes to history length alone.

V. CONCLUSION

We evaluated learned dynamics for MPPI-based tractor-trailer control against a Fiala-based analytic prior and additional controller baselines. In the main tracking experiments, learned MPPI achieved mean speeds closer to the target magnitudes and lower RMS hitch angles than analytic MPPI at every tested setpoint. It completed 62 of 72 reverse runs, compared with 41 for analytic MPPI, with the difference accounted for by fewer stalls. This advantage persisted on the tested slope and banking profiles, which were absent from training, although combined snow and terrain variation still limited reverse completion. On a separate Snow + Slope segment, the selected $H = 4$ and $H = 8$ predictors completed all reverse runs, but only $H = 8$ completed all forward runs. Matched training comparisons are needed to isolate the effect of history length. Future work will also evaluate trailer payload variation.

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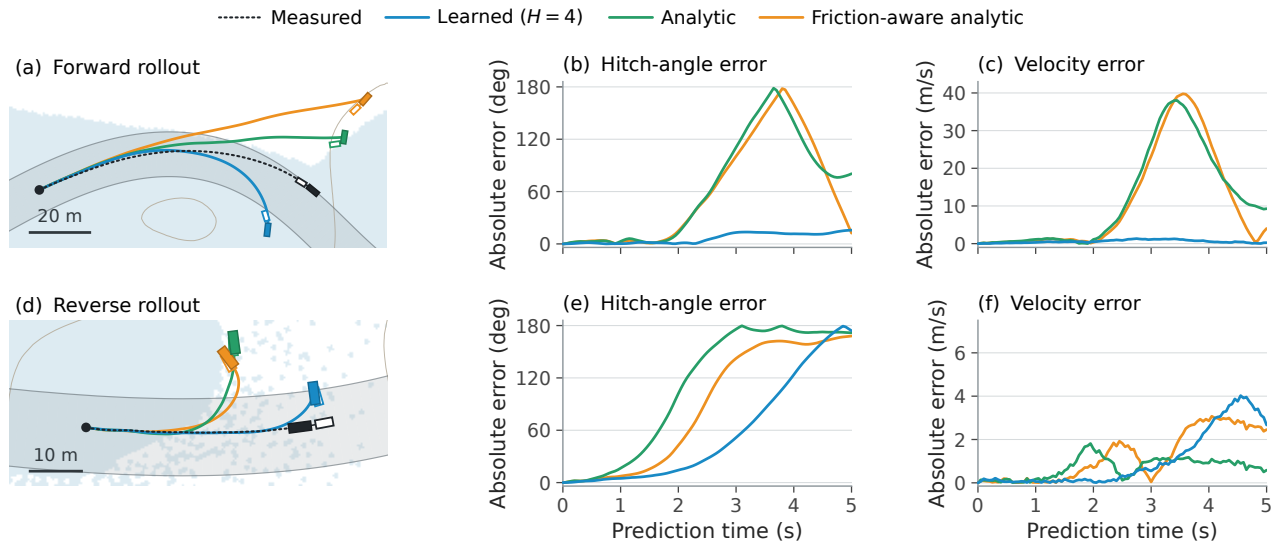


Fig. 5: Five-second open-loop dynamics predictions in Snow + Slope for forward (a,b,c) and reverse (d,e,f) driving. The measured trajectory is shown in dotted black, and the dynamics models' open-loop predictions are shown in solid color. Terminal tractor and trailer poses are denoted by solid and hollow boxes, respectively. The blue shading denotes snow patches with decreased friction, and elevation contours are spaced at 5 m.

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